

Project Name: Dalil Al-Ahkam: Semantic Retrieval & Citation System for Juristic Hadith Texts

Course: Software Engineering -- Requirements Engineering Assignment

Team Members:

Name: Mohammed Abdulwahab | Abdulrahman Al-Hattab

System Overview:

Dalil Al-Ahkam is a web-based academic retrieval platform designed for Sharia and Hadith researchers. The system matches natural-language juristic queries to authentic Prophetic Hadiths within accredited legal-evidence collections (*Ahadith Al-Ahkam*), providing exact academic citations and direct links to verified physical page scans without generating synthetic rulings or legal interpretations.

1. User Authentication & Session Management

1.1 The system shall allow users to register and authenticate using an email address and password or via Google authentication (OAuth2).

(Rationale: Providing standard email registration alongside Google OAuth2 simplifies user onboarding and secures user identities.)

1.2 The system shall enforce Time-based One-Time Password (TOTP) Two-Factor Authentication (2FA) for administrator accounts.

(Rationale: Multi-factor authentication prevents unauthorized administrative access to critical system configurations and source databases.)

1.3 The system shall allow the administrator to suspend or revoke user accounts at any time.

(Rationale: Administrative governance ensures platform integrity and prevents system abuse.)

2. Source & Reference Management

2.1 The system shall restrict indexed source texts exclusively to legal-evidence Hadith collections (*Ahadith Al-Ahkam*).

(Rationale: Limiting the corpus to juristic Hadith reference works guarantees contextual precision for legal research and eliminates irrelevant narrative texts.)

2.2 The system shall record and preserve academic edition metadata for every indexed book, including editor/verifier name (*Al-Muhaqqiq*), publication edition, publisher, and volume count.

(Rationale: Academic research standards in Sharia disciplines mandate citation from verified, accredited scholarly editions.)

2.3 The system shall allow the administrator to activate, suspend, or update indexed Hadith editions without interrupting system availability.

(Rationale: Maintaining and updating source datasets must occur seamlessly without taking the research platform offline.)

2.4 The system should allow users to submit formal requests for indexing new legal-evidence Hadith source books.

(Rationale: User feedback helps administrators identify and prioritize high-demand reference collections for future system releases.)

3. Juristic Query Processing & Retrieval

3.1 The system shall allow users to submit juristic research queries in free-form natural language without requiring exact Hadith keywords.

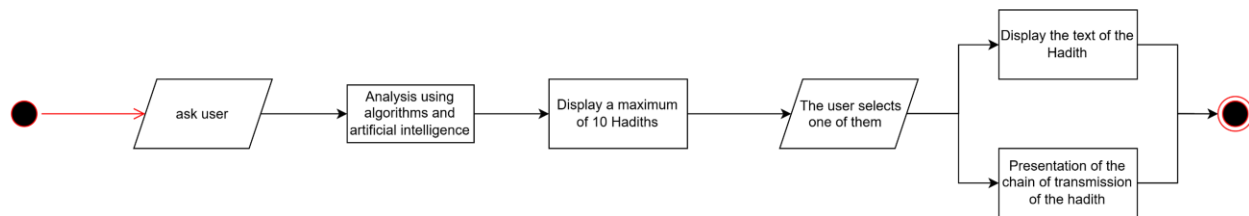
(Rationale: Researchers frequently query using conceptual legal terminology or contemporary phrasing rather than memorized classical wording.)

3.2 The system should allow users to restrict their query to a specific indexed source book or search across all indexed collections simultaneously.

(Rationale: Targeted querying enables comparative juristic analysis and reduces irrelevant results across multiple reference collections.)

3.3 The system should prompt the user for query clarification when a submitted search term is overly broad or linguistically ambiguous.

(Rationale: Clarifying researcher intent narrows the scope of retrieved texts and optimizes computational processing.)



3.4 The system shall retrieve relevant source texts directly without generating synthetic legal rulings, Fatwas, or automated juristic conclusions.

(Rationale: Academic and religious integrity strictly prohibits automated systems from deducing religious verdicts or introducing synthetic textual interpretations.)

4. Result Presentation & Academic Citation

4.1 The system shall retrieve and rank the top matching Hadiths (up to a maximum of 10) based on semantic relevance scores.

(Rationale: Ranking and limiting returned results prevents information overload and highlights the most legally relevant evidentiary texts.)

4.2 The system shall display verified academic citation metadata for each retrieved Hadith, including book title, verified scholarly edition, volume number, physical page number, and Hadith number.

(Rationale: Standardized and complete citation details are mandatory for formal thesis documentation and scholarly verification.)

4.3 The system shall provide a direct visual link to the digitized physical page scan (PDF) corresponding to each retrieved Hadith.

(Rationale: Researchers require visual confirmation of the printed source to inspect surrounding textual context and original marginalia.)

4.4 The system should allow users to export the retrieved Hadith, complete citation metadata, and scholarly evaluation in formatted text or PDF.

(Rationale: Standardized citation exports facilitate direct integration into academic research software, bibliographies, and paper footnotes.)

5. Transmission Chain (Sanad) & Scholarly Authenticity

5.1 The system shall display the transmission chain (Sanad) of the Hadith as documented in the source text, with an optional visual chain representation.

(Rationale: Visualizing the recorded sequence of narrators facilitates rapid inspection of chain continuity without automated interference.)

5.2 The system should display the authenticity rating (Hukm) and chain classification solely as static citations attributed to accredited Hadith scholars, without performing algorithmic chain evaluation.

(Rationale: Hadith criticism is an interpretive scholarly discipline with documented differences that cannot be automated without risking doctrinal errors.)

5.3 The system should display alternative scholarly evaluations when significant differences exist across accredited reference works for the same Hadith.

(Rationale: Documenting differing scholarly opinions provides comprehensive evidence for advanced academic research.)

6. User Workspace & Reporting

6.1 The system should allow authenticated users to save retrieved Hadiths and organize citations into custom research folders.

(Rationale: Researchers need to store and categorize evidentiary texts across multiple sessions during long-term research preparation.)

6.2 The system shall allow users to rate the relevance of retrieved Hadiths and submit detailed reports regarding inaccurate results, textual errors, or page citation discrepancies.

(Rationale: Direct user reporting ensures continuous data auditing and rapid correction of digitization or retrieval discrepancies.)

6.3 The system should allow users to view their previous search queries and access retrieved results from past sessions.

(Rationale: Retaining search history facilitates research continuity and eliminates redundant query processing.)

7. Non-Functional Requirements (Quality Constraints)

7.1 Performance: The system shall return query results within 3 seconds under normal operating conditions.

(Rationale: Fast response times maintain an efficient research workflow and prevent session abandonment.)

7.2 Security: The system shall ensure user search queries and saved research folders remain confidential and accessible only by the authenticated account owner.

(Rationale: Preserving academic privacy ensures intellectual property protection for researchers working on unpublished scholarly topics.)

7.3 Availability: The system shall maintain an operational availability rate of at least 99.99% (two minutes per day) during academic semesters.

(Rationale: Consistent platform availability prevents disruption to students and faculty during active research and study periods.)